ISM 6419 Data Visualization

FINAL PROJECT REPORT

Analyzing Electric Vehicle Adoption and Legislative Impact in Washington State

Course Instructor

Prof. Johannes Reichgelt

- **By**
- **Shrimanth Kavuluru**

  U88472896

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Introduction:

This project looks at electric vehicle (EV) trends in Washington State by combining data from various sources, like EV registrations, charging stations, tax breaks, and income levels. It aims to understand what factors influence EV adoption and how charging stations are used. The goal is to provide useful insights that can help shape future policies and strategies, making it easier for people to adopt EVs and improving charging infrastructure across the state.

Background of the Project:

Electric vehicles (EVs) play a crucial role in reducing greenhouse gas emissions and advancing sustainable transportation. Washington State has been at the forefront of promoting EV adoption through progressive policies like the 2019 HB 2042 bill, which offers tax exemptions and incentives to encourage consumers to switch to cleaner alternatives. Despite these efforts, understanding the nuances of EV adoption, the influence of legislative measures, and the factors driving consumer choices requires a detailed and comprehensive analysis.

Ambitiousness of the Project:

This ambitious project analyzes electric vehicle (EV) trends in Washington State by integrating data from various sources, including EV population, registration activities, charging station availability, tax exemptions, and county income levels. Key highlights of the project include:

Data Integration and Cleaning

Combining Diverse Data Sources: Merging and cleaning data from multiple sources ensures consistency and accuracy, crucial for reliable insights.

Advanced Data Visualization: Utilizes heat maps, time-series analyses, and bar charts to present findings clearly and engagingly.

Evaluating Legislative Impact: Assesses the influence of laws like the 2019 HB 2042 bill on EV adoption, providing insights into policy effectiveness.

Addressing Multiple Research Questions:

Comprehensive Approach: Explores various aspects of EV adoption, including geographic distribution, economic factors, charging station availability, and tax exemptions.

Application & Insights: Aims to deliver insights to shape future policies and strategies, promoting sustainable transportation and enhancing EV infrastructure.

These efforts help policymakers, industry stakeholders, and the public make informed decisions to boost EV adoption and infrastructure in Washington State.

Research Questions:

- **Research Question 1: How has the geographic distribution of electric vehicles evolved in Washington State over the past five years?**
- **Datasets:** Electric Vehicle Population Data
  Electric Vehicle Title and Registration Activity

  Objective: To use the population data to map the distribution of electric vehicles and track changes over time using the registration activity data. This can help visualize adoption rates and identify regions with higher or lower adoption

- **Research Question 2: How does the availability of charging stations influence the adoption of electric vehicles in Washington State?**
  - **Datasets:** Electric Vehicle Population Data
  - Alternative Fuels Data Center

  Objective: To analyze the availability of charging stations for Electric Vehicles across the county and observe what actions would be necessary for enhancing the situation.

- **Research Question 3: What are the economic factors influencing the adoption of electric vehicles, such as sale price and base MSRP, and how do they vary across different regions?**

  Datasets: Electric Vehicle Title and Registration Activity

  Personal Income by County and Metropolitan Area, 2022

  Objective: Observing the trend of the proportion of registered vehicles under the influence of base and sale price.

- **Research Question 4: How does the eligibility for tax exemptions influence the choice of electric vehicle models among consumers?**
- **Datasets:**WA Tax Exemptions - Potential Eligibility by Make/Model

  Electric Vehicle Title and Registration Activity

  Objective: To analyze how tax exemption eligibility influences the choice of electric vehicle models among consumers in Washington State .

- **Research Question 5: How do economic factors and the availability of fuel stations (including electric vehicle charging stations) influence the adoption of electric vehicles in Washington State?**
- **Datasets Used:** Electric Vehicle Population Data
  Alternative Fuels Data Center
  Personal Income by County and Metropolitan Area, 2022
- Objective: To understand how economic factors and the availability of fuel stations, including EV charging stations, influence the adoption of electric vehicles in Washington State.

- **Research Question 6: How do geographic and economic factors influence the efficiency and utilization of electric vehicle charging infrastructure in Washington State?**
- **Datasets:** Electric Vehicle Population Data
- Alternative Fuels Data Center
- Income by County and Metropolitan Area, 2022

  Objective: The objective is to analyze how geographic and economic factors influence the utilization of electric vehicle charging infrastructure in Washington State.

Data Source

The data for this project is sourced from multiple government sources as mentioned below**.** It includes datasets on electric vehicle population, title registration activity, and tax exemption eligibility, alternative fuel stations availability and income data providing a comprehensive basis for analyzing EV adoption and legislative impacts in Washington State.

Methodology:

I have linked 5 data sets that have been extracted from United States Government Sources.

1)Electric Vehicle Population Data

Columns:VIN (1-10) County City State Postal Code Model Year Make Model Electric Vehicle Type Clean Alternative Fuel Vehicle (CAFV) Eligibility Electric Range Base MSRP Legislative District DOL Vehicle ID Vehicle Location Electric Utility 2020 Census Tract.

2)Alternative Fuels Data Center

Columns: Fuel Type Code Station Name Street Address Intersection Directions City State ZIP Plus4 Station Phone Status Code Expected Date Groups With Access Code Access Days Time Cards Accepted BD Blends NG Fill Type Code NG PSI EV Level1 EVSE Num EV Level2 EVSE Num EV DC Fast Count EV Other Info EV Network EV Network Web Geocode Status Latitude Longitude Date Last Confirmed ID Updated At Owner Type Code Federal Agency ID Federal Agency Name Open Date Hydrogen Status Link NG Vehicle Class LPG Primary E85 Blender Pump EV Connector Types Country Intersection Directions (French) Access Days Time (French) BD Blends (French) Groups With Access Code (French) Hydrogen Is Retail Access Code Access Detail Code Federal Agency Code Facility Type CNG Dispenser Num CNG On-Site Renewable Source CNG Total Compression Capacity CNG Storage Capacity LNG On-Site Renewable Source E85 Other Ethanol Blends EV Pricing EV Pricing (French) LPG Nozzle Types Hydrogen Pressures Hydrogen Standards CNG Fill Type Code CNG PSI CNG Vehicle Class LNG Vehicle Class EV On-Site Renewable Source Restricted Access RD Blends RD Blends (French) RD Blended with Biodiesel RD Maximum Biodiesel Level NPS Unit Name CNG Station Sells Renewable Natural Gas LNG Station Sells Renewable Natural Gas Maximum Vehicle Class EV Workplace Charging Federal Funding Types

3)WA Tax Exemptions - Potential Eligibility by Make/Model Excluding Vehicle Price Criteria

Columns: Model Year Make Vehicle Model Description Alternative Fuel Type

4)Personal Income by County and Metropolitan Area

Columns
Table 1. Per Capita Personal Income, by County, 2020–2022

Per capita personal income1 Percent change from preceding period
Dollars Rank in state Percent change Rank in state
2020 2021 2022 2022 2021 2022 2022

5)Electric Vehicle Title and Registration Activity

Columns: Clean Alternative Fuel Vehicle Type VIN (1-10) DOL Vehicle ID Model Year Make Model Primary Use Electric Range Odometer Reading Odometer Reading Description New or Used Vehicle Sale Price Sale Date Base MSRP Transaction Type Transaction Date Year County City State Postal Code 2019 HB 2042: Clean Alternative Fuel Vehicle (CAFV) Eligibility Meets 2019 HB 2042 Electric Range Requirement Meets 2019 HB 2042 Sale Date Requirement Meets 2019 HB 2042 Sale Price/Value Requirement 2019 HB 2042: Battery Range Requirement 2019 HB 2042: Purchase Date Requirement 2019 HB 2042: Sale Price/Value Requirement Electric Vehicle Fee Paid Transportation Electrification Fee Paid Hybrid Vehicle Electrification Fee Paid 2020 GEOID Legislative District Electric Utility

Data Pre-Processing:

Data preprocessing is essential to ensure that the datasets are clean, consistent, and ready for analysis. Preprocessing steps that has been followed for the provided datasets:

Data Cleaning:

Identifying and handling missing values: Check for missing values in the datasets and decide on an appropriate strategy for handling them, such as imputation or removal. The introduction sheets were discarded.

Removing duplicates: Removed duplicate entries from the datasets to avoid redundancy in the analysis.

Deleting Unnecessary columns to avoid confusion. Some of the data sets like Alternative fuel dataset and EV Registrations data set are to be cleaned and normalized rigorously as they had numerous column and information which is not required for the project objective.

Data Transformation:

Normalization: Standardized numerical features such as EV prices, income levels, and charging station counts to a common scale for improved comparison and analysis.

Categorical Encoding: Converted categorical data, such as EV model types and geographic regions appropriate formats.

Aggregation: Compiled data at various geographic levels (e.g., county, city) to examine regional trends and patterns in EV adoption and charging infrastructure utilization.

Data Formatting and Consistency:

Ensuring that the data is formatted correctly for visualization analysis, such as converting dates (YEAR) to a standardized format and ensuring consistency in variable naming conventions. Applied consistent naming conventions for columns to ensure clarity and ease of analysis (e.g., EV_Count, Income_Level, Charging_Station_Count).

EV Adoption Analysis: Cleaned and standardized data on EV prices, charging stations, and regional economic factors for comprehensive analysis.

Economic Factors: Normalized income data across counties to accurately compare EV adoption rates.

Tax Exemptions and EV Models: Encoded tax exemption eligibility and analyzed its correlation with EV model popularity.

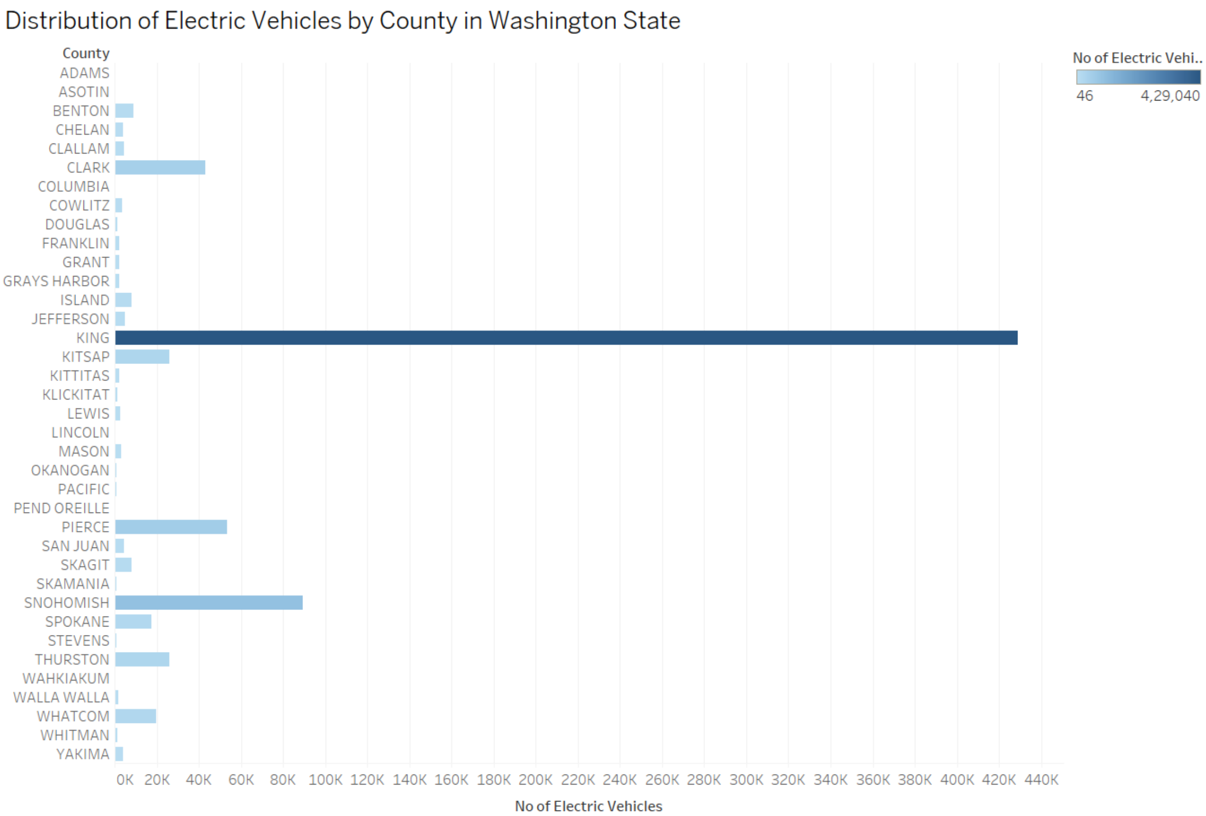
Infrastructure Placement: Aggregated and analyzed charging station data to assess the impact of strategic placement on utilization.

Data Visualization Tool:

Following the observations and understanding from the course , I have also followed a approach towards my final project where I preferred tableau for all the Data integration and visualization considering its advanced and customizable visualization which lead to quick interactive dashboards. For the steps of cleaning & transforming the available data I used R & Python accordingly for the preprocessing.

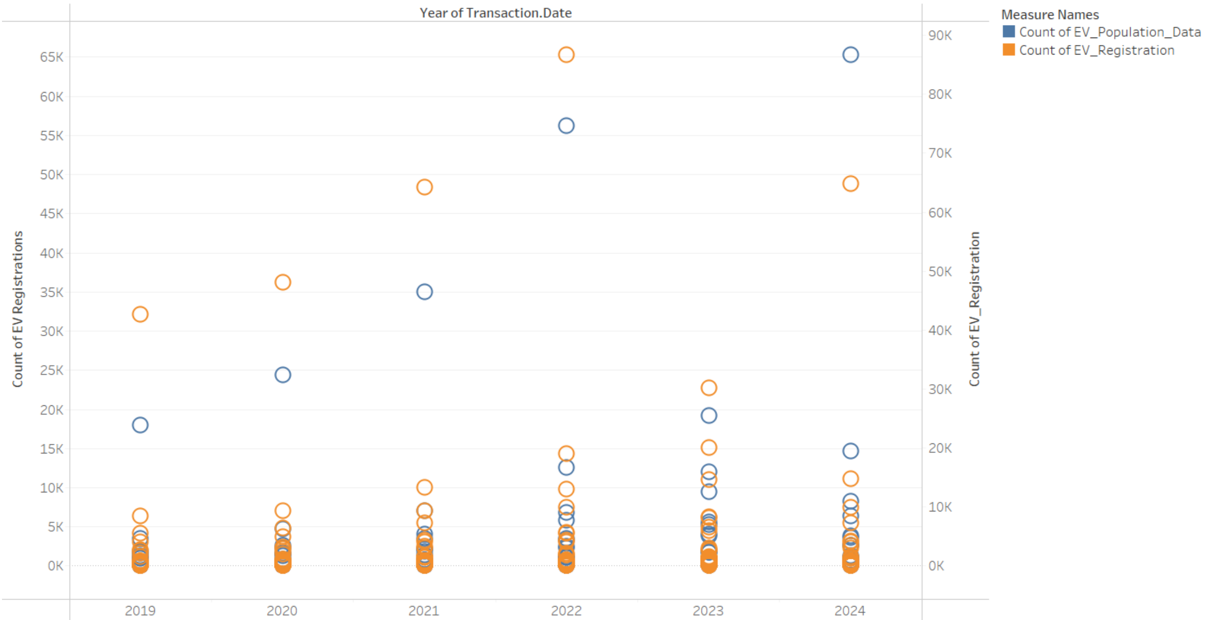
ANALYSIS

- How has the geographic distribution of electric vehicles evolved in Washington State over the past five years?
- How does the availability of charging stations influence the adoption of electric vehicles in Washington State?



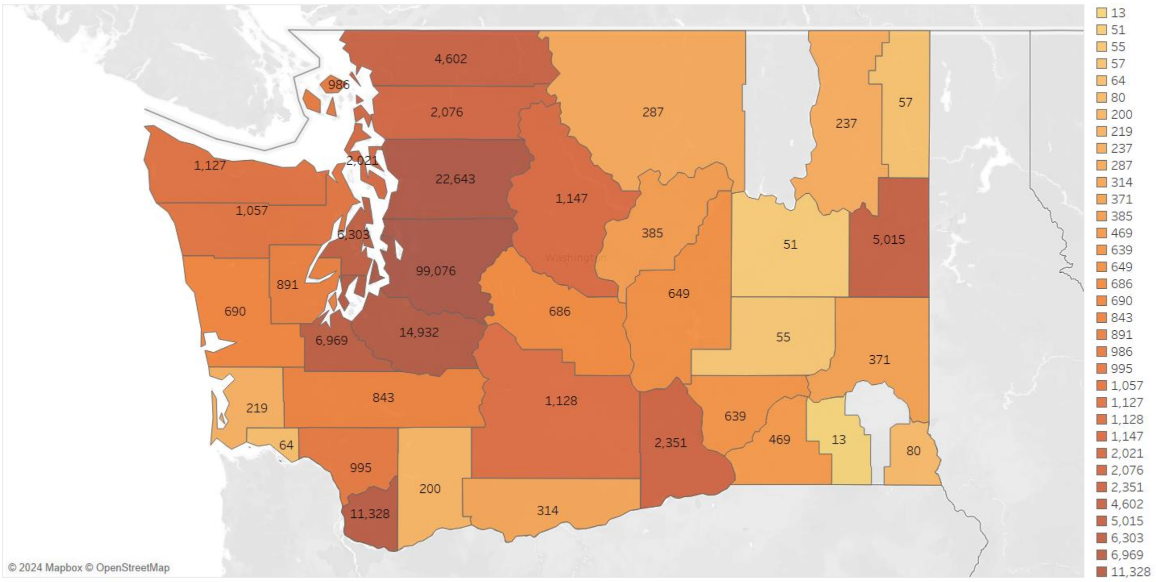
No of Electric Vehicles for each County. Color shows No of Electric Vehicles. The data is filtered on State, which keeps WA.

Electric Vehicle Registration Count by County and Year



Count of EV_Population_Data and count of EV_Registration for each Transaction.Date Year. Color shows details about count of EV_Population_Data and count of EV_Registration. Details are shown for COUNTY. The data is filtered on Transaction.Date Year and count of EV_WA_TaxExemption. The Transaction.Date Year filter keeps 6 members. The count of EV_WA_TaxExemption filter ranges from 9 to 759. The view is filtered on count of EV_Population_Data and count of EV_Registration. The count of EV_Population_Data filter ranges from 1 to 83,432. The count of EV_Registration filter ranges from 1 to 1,22,001.

EV Population Heat Map



Map based on Longitude (generated) and Latitude (generated). Color shows details about count of EV_Population_Data. The marks are labeled by count of EV_Population_Data. Details are shown for County. The data is filtered on State and POSTAL.CODE (Cleaned Electric Vehicle Registration.csv). The State filter keeps WA. The POSTAL.CODE (Cleaned Electric Vehicle Registration.csv) filter keeps 548 members. The view is filtered on County, which keeps 39 members.

Analyzing the above bar chart it is seen that King county has highest number 420000 electric vehicles and the distribution of EV over the county's is uneven. Other counties like Snohomish, Pierce, and Clark have noticeable numbers, though far less than King County.

From the Scatter plot we can observe a steady increase in EV registrations from 2019 to 2024, and continuing like above King county has the highest registration.

To promote broader adoption across Washington State, efforts should focus on expanding charging infrastructure in underrepresented areas, particularly in rural and eastern counties.

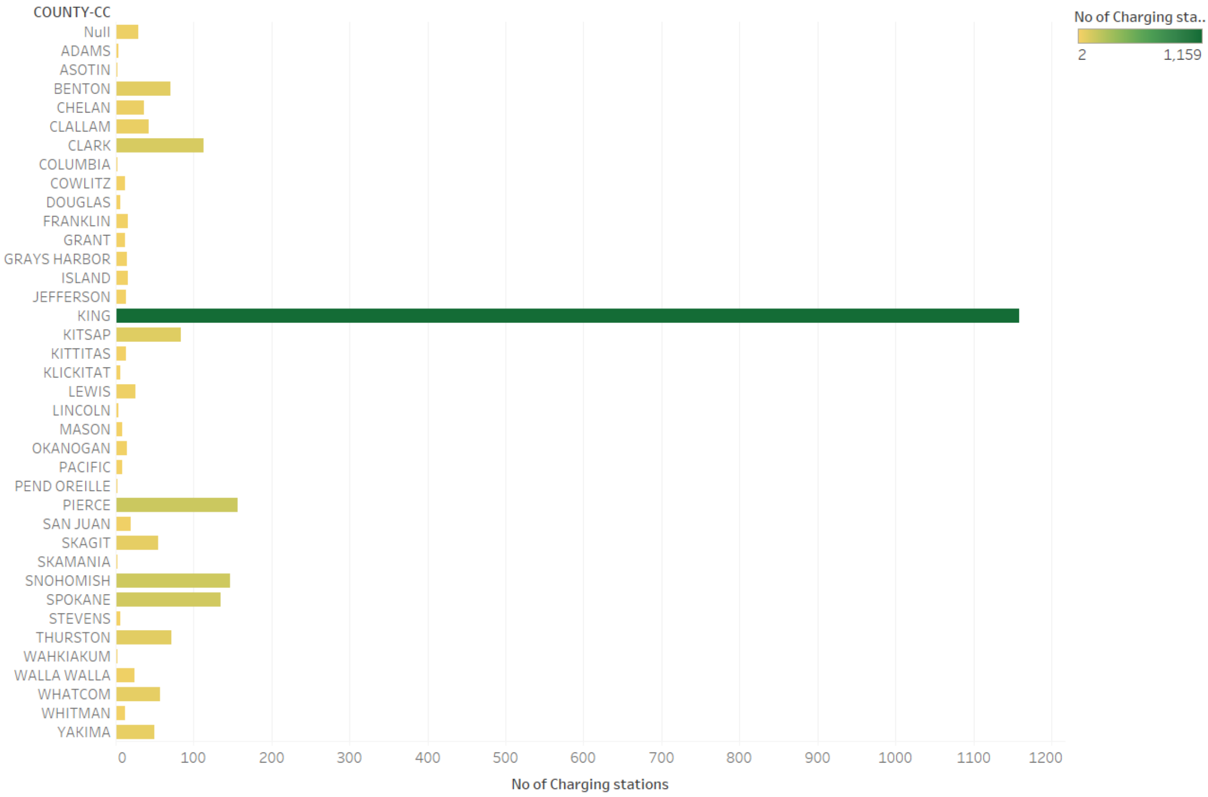
Shrimanth Kavuluru U88472896

This strategic expansion could help balance the geographic distribution of electric vehicles and support the state's overall sustainability goals.

5. How do economic factors and the availability of fuel stations (including electric vehicle charging stations) influence the adoption of electric vehicles in Washington State?

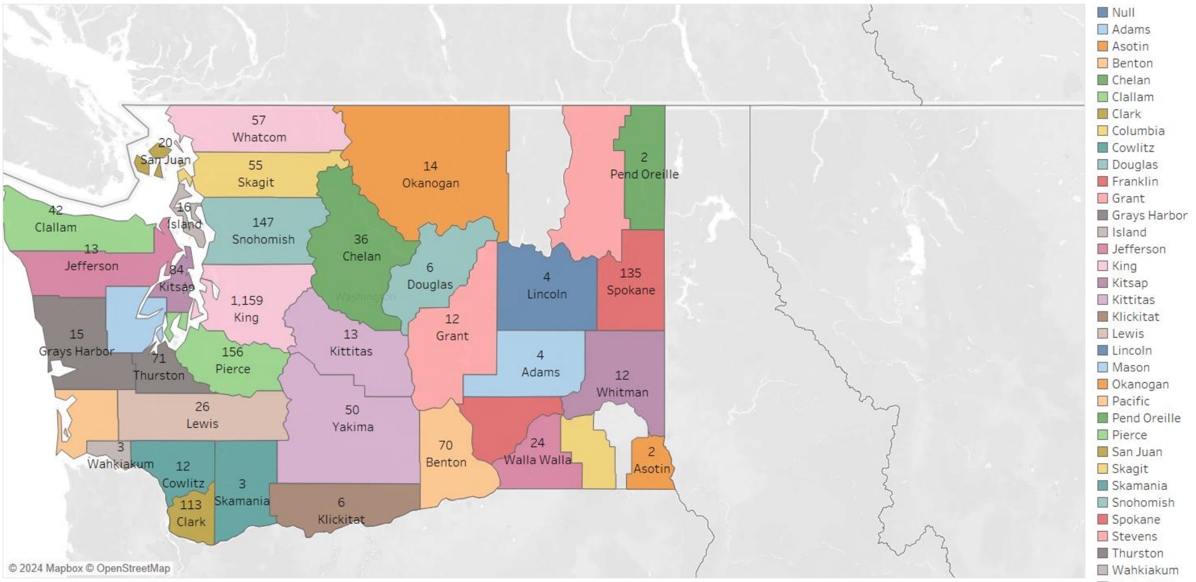
6. How do geographic and economic factors influence the efficiency and utilization of electric vehicle charging infrastructure in Washington State?

Distribution of Charging Stations by County in Washington State



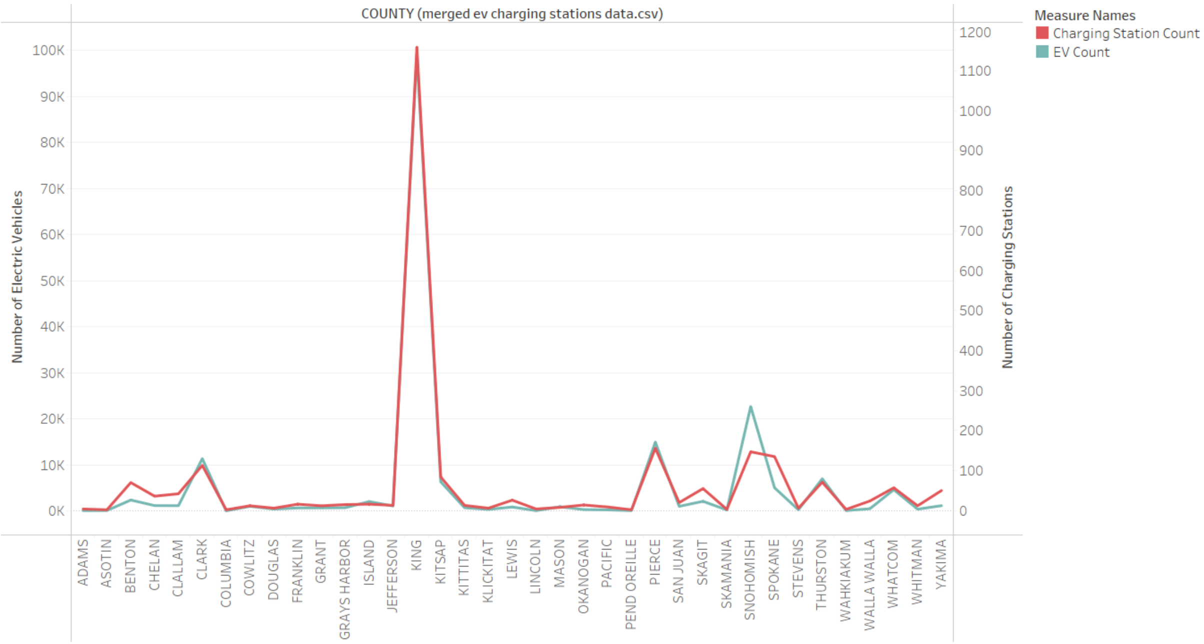
No of Charging stations for each COUNTY-CC. Color shows No of Charging stations.

Distribution of Electric Vehicles, Charging Stations, and Income by County in Washington State



Map based on Longitude (generated) and Latitude (generated). Color shows details about County1. The marks are labeled by No of Charging stations and County1. Details are shown for County1 and Measure Names.

Distribution of Electric Vehicles and Charging Stations by County in Washington State

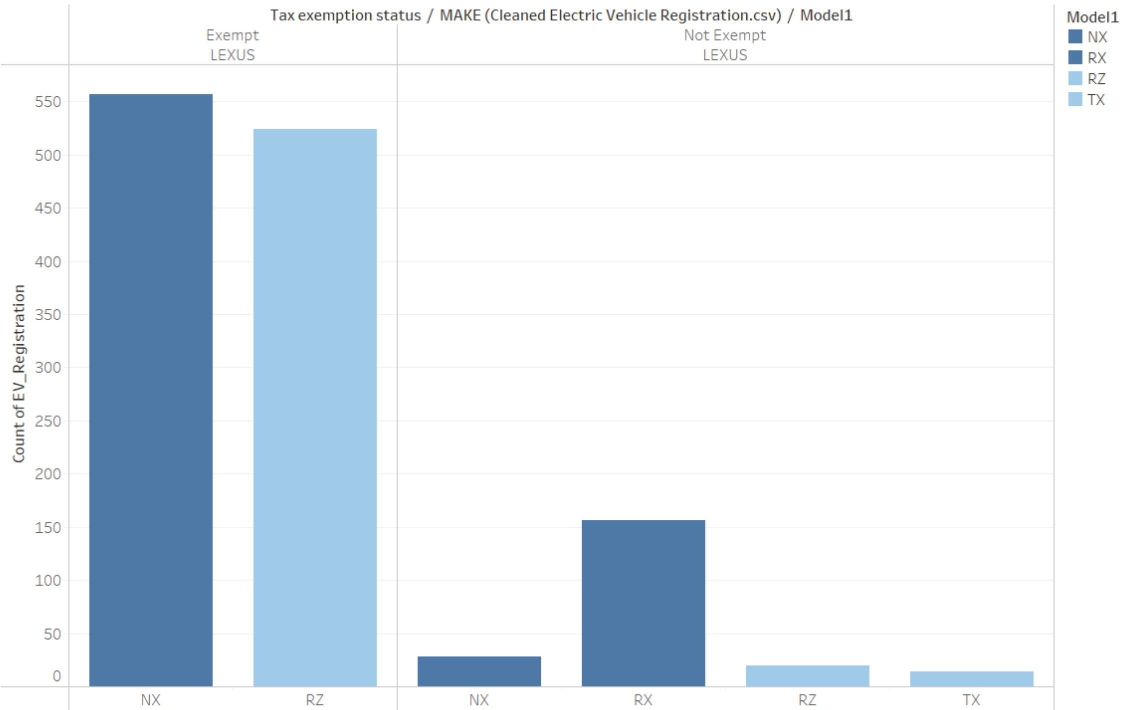


The trends of EV Count and Charging Station Count for COUNTY (merged ev charging stations data.csv). Color shows details about EV Count and Charging Station Count.

3. What are the economic factors influencing the adoption of electric vehicles, such as sale price and base MSRP, and how do they vary across different regions?

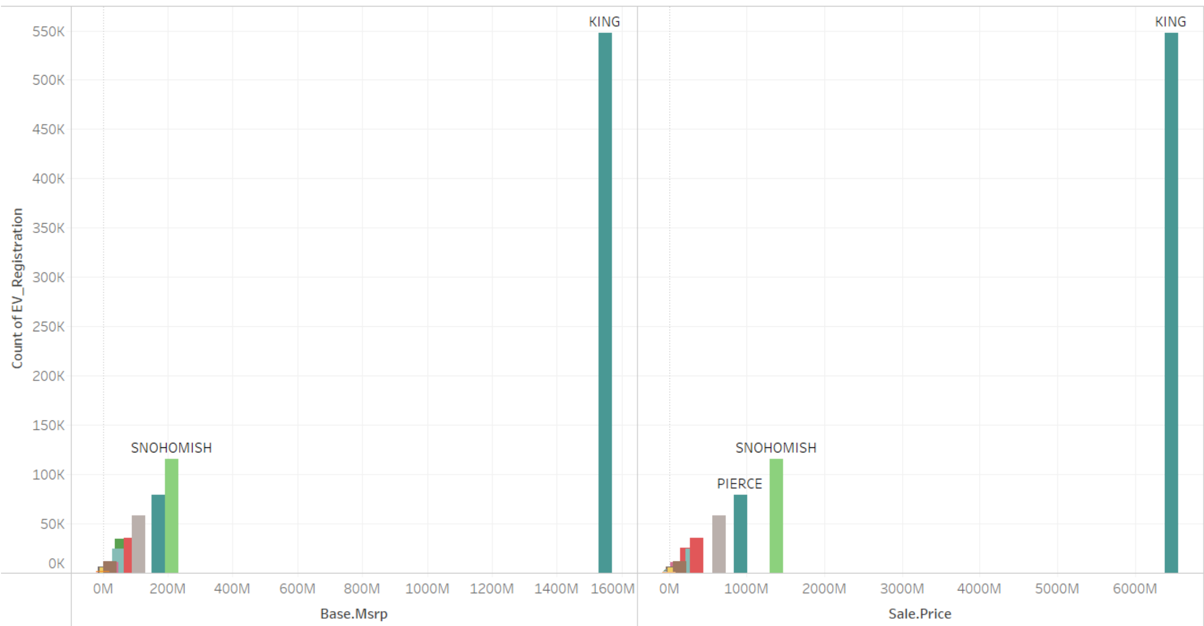
4. How does the eligibility for tax exemptions influence the choice of electric vehicle models among consumers?

EV vs Tax Exemption



Count of EV_Registration for each Model1 broken down by Tax exemption status and MAKE (Cleaned Electric Vehicle Registration.csv). Color shows details about Model1. The view is filtered on Tax exemption status, count of EV_Registration and MAKE (Cleaned Electric Vehicle Registration.csv). The Tax exemption status filter excludes Null. The count of EV_Registration filter includes everything. The MAKE (Cleaned Electric Vehicle Registration.csv) filter keeps LEXUS.

No of EV vs Base Price



Sum of Base.Msrp and sum of Sale.Price vs. count of EV_Registration. Color shows details about COUNTY. The marks are labeled by COUNTY. Details are shown for COUNTY (Cleaned Electric Vehicle Registration.csv).

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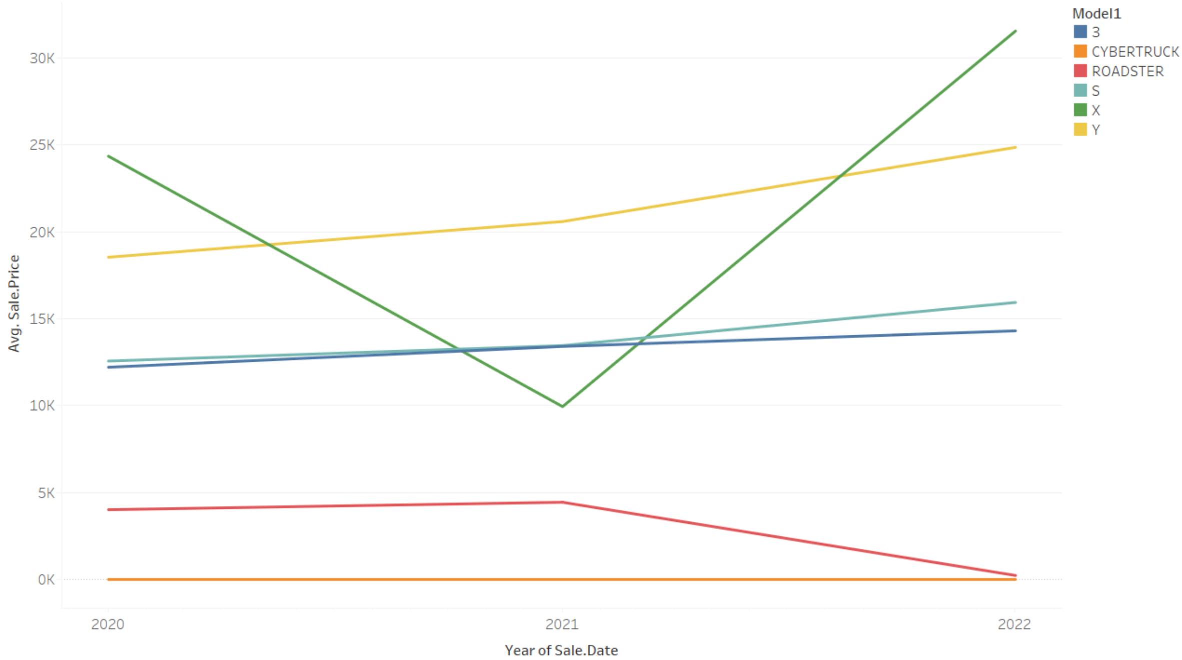
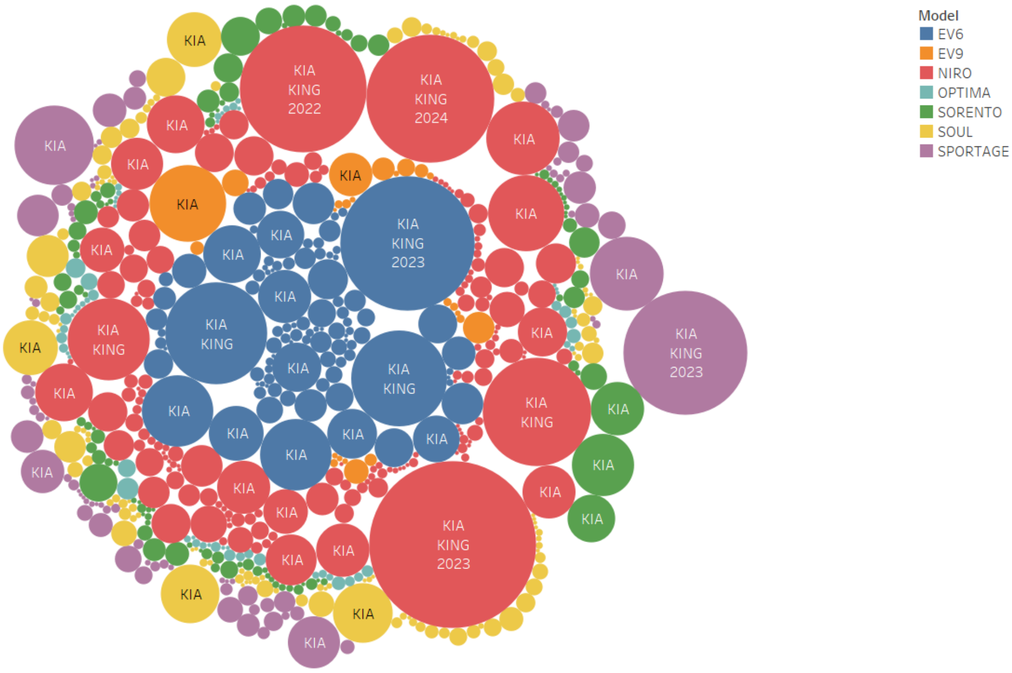
Trends in Average Sale Price of Electric Vehicle Models Over Time



The trend of average of Sale.Price for Sale.Date Year. Color shows details about Model1. The data is filtered on Sale.Date Year and MAKE (Cleaned Electric Vehicle Registration.csv). The Sale.Date Year filter keeps 2020, 2021 and 2022. The MAKE (Cleaned Electric Vehicle Registration.csv) filter keeps TESLA.

Overview of Electric Vehicle Records by Make, Model, County, and Tax Exemption Status



Make, COUNTY (Cleaned Electric Vehicle Registration.csv) and Transaction.Date Year. Color shows details about Model. Size shows No of Electric Vehicles. The marks are labeled by Make, COUNTY (Cleaned Electric Vehicle Registration.csv) and Transaction.Date Year. Details are shown for Tax exemption status. The data is filtered on Sale.Date Year, which keeps 2020, 2021, 2022, 2023 and 2024. The view is filtered on Transaction.Date Year, No of Electric Vehicles and Make. The Transaction.Date Year filter keeps 6 members. The No of Electric Vehicles filter keeps non-Null values only. The Make filter keeps KIA.

Shrimanth Kavuluru U88472896

Average Sale and Base Price Difference of EV Make and model by County

Average of PRICE DIFFERENCE for each Model1. Color shows details about COUNTY (Cleaned Electric Vehicle Registration.csv). Details are shown for COUNTY (Cleaned Electric Vehicle Registration.csv). The data is filtered on STATE (Cleaned Electric Vehicle Registration.csv) and Make. The STATE (Cleaned Electric Vehicle Registration.csv) filter keeps WA. The Make filter keeps AUDI. The view is filtered on COUNTY (Cleaned Electric Vehicle Registration.csv), which keeps 33 members.

From the above visualizations , we can observe

- King County has the highest number of electric vehicles and charging stations, driving significant EV adoption.
- Urban areas show higher EV concentrations, with notable growth in registrations over the past two years.
- Availability of charging stations correlates with higher EV adoption rates in counties like Snohomish and Pierce.
- Higher income regions, such as King and Snohomish, have more EV registrations, influenced by affordability and price trends.
- Increasing sale prices of Tesla models and tax exemptions for Lexus models impact consumer choices and adoption rates.
- These findings can guide policies to enhance EV infrastructure and promote sustainable transportation statewide.

Key Findings:

1)Geographic Distribution Of EV's

We can observe tfrom the visualizations that the distribution of EV's is much higher areas like King County etc, which apparently translate to urban regio regions, so EV's are comparatively very low in country side. The reason for this might be lacking infrastructure and accessibility.

2)Temporal Trends in EV

Over the last couple of years, the EV registrations have increased rapidly with notable spikes in certain years. As said above though this adoption rate is high in urban & suburban areas, it indicates growing acceptance of EV.

3)Availability of Charging Stations & Utilization

There is a direct correlation between the number of charging stations and the adoption of EVs. Counties with more charging stations tend to have higher numbers of EV registrations.

Charging stations are used more in counties with high EV populations.Wealthier counties have more charging stations and use them more efficiently due to higher income levels.

Some counties with low EV adoption rates also have fewer charging stations, highlighting a potential area for infrastructure development.

4) Economic Factors & EV prices

The average sale price (SP) and base MSRP of electric vehicles (EVs) vary across different counties, with wealthier counties more commonly featuring higher-priced EVs. There is a clear trend indicating that regions with higher per capita income have higher rates of EV adoption, suggesting that economic capability plays a significant role in the adoption of EVs.

5)Impact of Tax Exemptions

Tax exemptions greatly affect which EV models consumers choose. Models that qualify for tax exemptions are adopted more often. In counties where people are more aware of and have easier access to these tax breaks, consumers are more likely to pick EV models that come with these financial benefits.

6)Adoption vs Base Price:

A comparative analysis of the number of EVs versus their base price shows that mid-range priced EVs have higher adoption rates.

High-end luxury EVs are less common but show steady sales in affluent counties.

7)Impact of Economic and Geographic Factors:

Geographic and economic factors jointly impact the efficiency and usage of EV charging infrastructure. Regions with stronger economies and strategically placed charging stations experience higher utilization and greater satisfaction among EV users.

Conclusion:

The project looks at the factors that affect electric vehicle (EV) adoption in Washington State. We can identify that geography and economy are crucial things in the Electric Vehicle sector. Cities with higher incomes and better charging infrastructure have more EVs. Tax breaks acts as an effective motivator for consumers when choosing EV models. The availability and smart placement of charging stations are essential for increasing EV use, especially in rural and less wealthy areas.

The study highlights the need for targeted infrastructure, economic incentives, and public awareness campaigns to boost EV adoption across different regions. By addressing gaps in charging station availability and using economic incentives, policymakers can improve EV adoption and efficiency, supporting sustainable transportation goals.

These findings can help shape future policies and initiatives to speed up EV adoption, ensure fair access to EV technology, and optimize the rollout of charging infrastructure across Washington State.